

COMPREHENSIVE SOC / NOC PROJECT DOCUMENTATION

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Abstract

Building a Hybrid Enterprise Operations Lab for Threat Detection and Infrastructure
Monitoring

Date: September 2026

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1. Overview

This project is a fully functional, hybrid NOC + SOC lab that reflects how real enterprises monitor both infrastructure, health and security posture from a single operations environment. It combines centralized SIEM-based threat detection with performance/uptime monitoring, all running on a segmented network built with enterprise-grade firewall routing.

The lab simulates real attack scenarios - reconnaissance, brute force, privilege escalation, and web application attacks - and validates that the detection stack actually catches them, with incident handling modeled on NIST SP 800-61 Rev 2 and detections mapped to the MITRE ATT&CK framework.

2. Executive Summary

This project establishes a fully functional, hybrid Operations Center environment that combines both Network Operations Center (NOC) and Security Operations Center (SOC) capabilities. Built as a comprehensive training and simulation lab, it demonstrates the ability to monitor IT infrastructure health while simultaneously detecting, analyzing, and responding to cyber security threats. The environment is designed to reflect real-world enterprise setups, complete with secure network segmentation and centralized logging.

At its core, this project solves the dual challenge of visibility and security. By integrating advanced firewall routing, endpoint threat detection, and continuous performance monitoring, the lab provides complete oversight over both physical and virtual network assets. This allows administrators to quickly identify when a server goes offline or when an unauthorized user attempts to breach the network.

Throughout this project, various attack scenarios were simulated to validate the system's defenses. These ranged from internal unauthorized access attempts to external web application attacks. The successful detection and documentation of these events demonstrate the lab's effectiveness and its alignment with professional incident handling frameworks like NIST SP 800-61 Rev 2.

3. Project Objectives

This project was built to address the following key objectives:

- Establish a realistic, segmented network environment using enterprise-grade routing and firewall rules.

pfSense
COMMUNITY EDITION

System Interfaces Firewall Services VPN Status Diagnostics Help

Firewall / Rules / WAN

Floating **WAN** LAN CORP DMZ

Rules (Drag to Change Order)

	States	Protocol	Source	Port	Destination	Port	Gateway	Queue	Schedule	Description	Actions
☐	✓ 0/10.71 MIB	IPv4 TCP	*	*	10.10.30.102	80 - 443	*	none		NAT NAT WAN HTTP to DMZ Web	
☐	✓ 0/0 B	IPv4 TCP	*	*	10.10.10.50	8080	*	none		NAT Zabbix WebGUI Forward	

Add Add Delete Toggle Copy Save Separator

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System Interfaces Firewall Services VPN Status Diagnostics Help

Firewall / Rules / LAN

Floating WAN **LAN** CORP DMZ

Rules (Drag to Change Order)

	States	Protocol	Source	Port	Destination	Port	Gateway	Queue	Schedule	Description	Actions
☐	✓ 3/61.85 MIB	*	*	*	LAN Address	443 80	*	*		Anti-Lockout Rule	
☐	✓ 0/45.49 MIB	IPv4 UDP	LAN subnets	*	10.10.30.102	161 (SNMP)	*	none		Allow LAN to DMZ SNMP Monitoring	
☐	✓ 0/0 B	IPv4 TCP	LAN subnets	*	10.10.30.102	80 - 443	*	none		Allow LAN to DMZ Web Server	
☐	✓ 27/3.21 GIB	IPv4 *	LAN subnets	*	*	*	*	none		Default allow LAN to any rule	
☐	✓ 0/0 B	IPv4 *	LAN subnets	*	*	*	*	none		Default allow LAN IPv6 to any rule	

Add Add Delete Toggle Copy Save Separator

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System Interfaces Firewall Services VPN Status Diagnostics Help

Firewall / Rules / CORP

Floating WAN LAN **CORP** DMZ

Rules (Drag to Change Order)

	States	Protocol	Source	Port	Destination	Port	Gateway	Queue	Schedule	Description	Actions
☐	✓ 0/1 KiB	IPv4 ICMP any	CORP subnets	*	10.10.10.50	*	*	none		Allow ICMP to Wazuh Manager	
☐	✓ 1/1.51 MIB	IPv4 TCP/UDP	CORP subnets	*	10.10.10.50	Wazuh_Zabbix_Ports	*	none		Allow CORP to Wazuh & Zabbix Allowed Ports Only	
☐	✗ 0/5 KiB	IPv4 *	CORP subnets	*	10.10.10.0/24	*	*	none		Block CORP access to MGMT	
☐	✓ 5/1.34 GiB	IPv4 *	CORP subnets	*	*	*	*	none		Allow CORP Outbound to Internet	

Add Add Delete Toggle Copy Save Separator

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System ▾ Interfaces ▾ Firewall ▾ Services ▾ VPN ▾ Status ▾ Diagnostics ▾ Help ▾

Firewall / Rules / DMZ

Floating WAN LAN CORP **DMZ**

Rules (Drag to Change Order)

☐	States	Protocol	Source	Port	Destination	Port	Gateway	Queue	Schedule	Description	Actions
☐	✓ 1/1.81 MiB	IPv4 TCP/UDP	DMZ subnets	*	10.10.10.50	1514-1515	*	none		Allow DMZ to Wazuh Manager (1514-1515)	
☐	✓ 0/205 KiB	IPv4 TCP/UDP	DMZ subnets	*	DMZ address	53 (DNS)	*	none		Allow DMZ to local DNS	
☐	✗ 0/56 KiB	IPv4 *	DMZ subnets	*	10.0.0.0/8	*	*	none		Block DMZ from initiating connections to LAN and CORP	
☐	✓ 0/191.80 MiB	IPv4 *	DMZ subnets	*	*	*	*	none		Allow DMZ Outbound to Internet	

Add Add Delete Toggle Copy Save Separator

Interfaces Global Settings Updates **Alerts** Blocks Files Pass Lists Suppress Logs View Logs Mgmt SID Mgmt

Sync IP Lists

Alert Log View Settings

Instance to View: (WAN) WAN Edge Traffic Inspection
Choose which instance alerts you want to inspect.

Save or Remove Logs: Download Clear
All alert log files for selected interface will be downloaded. Clear the currently active Alerts log file.

Save Settings: Save ☒ Refresh
Save auto-refresh and view settings. Default is ON. Number of alerts to display. Default is 250.

Alert Log View Filter

Last 250 Alert Entries. (Most recent entries are listed first)

Date	Action	Pri	Proto	Class	Src	SPort	Dst	DPort	GID:SID	Description
09/17/2026 22:34:01	⚠	3	TCP	Not Suspicious Traffic	192.168.215.131	26452	91.189.91.82	80	1:2013504	ET INFO GNU/Linux APT User-Agent Outbound likely related to package management
09/17/2026 22:28:08	⚠	3	TCP	Not Suspicious Traffic	192.168.215.131	23162	91.189.91.83	80	1:2013504	ET INFO GNU/Linux APT User-Agent Outbound likely related to package management
09/17/2026 22:14:06	⚠	3	TCP	Not Suspicious Traffic	192.168.215.131	7966	185.125.190.82	80	1:2013504	ET INFO GNU/Linux APT User-Agent Outbound likely related to package management
09/17/2026 22:14:06	⚠	3	TCP	Not Suspicious Traffic	192.168.215.131	7966	185.125.190.82	80	1:2013504	ET INFO GNU/Linux APT User-Agent Outbound likely related to package management

- Deploy a centralized monitoring solution (NOC) to track server uptime, network performance, and resource availability.
- Implement a Security Information and Event Management (SIEM) system (SOC) to collect and analyze security logs from multiple endpoints.
- Validate threat detection capabilities by executing simulated attacks mapped to the MITRE ATT&CK framework.

- Develop troubleshooting skills by resolving integration and configuration issues across various platforms and operating systems.

4. Scope

In-Scope:

- Deployment and configuration of pfSense, Wazuh, Zabbix, and target endpoints (Windows 10, Ubuntu Server).
- Implementation of network segmentation (LAN, DMZ, CORP).
- Configuration of data ingestion from endpoints (Sysmon, Linux auth logs, Nginx access logs).
- Simulation of specific threat vectors (Reconnaissance, Privilege Escalation, Web Attacks, Brute Force).
- Creation of alert filters and dashboards.

Out-of-Scope:

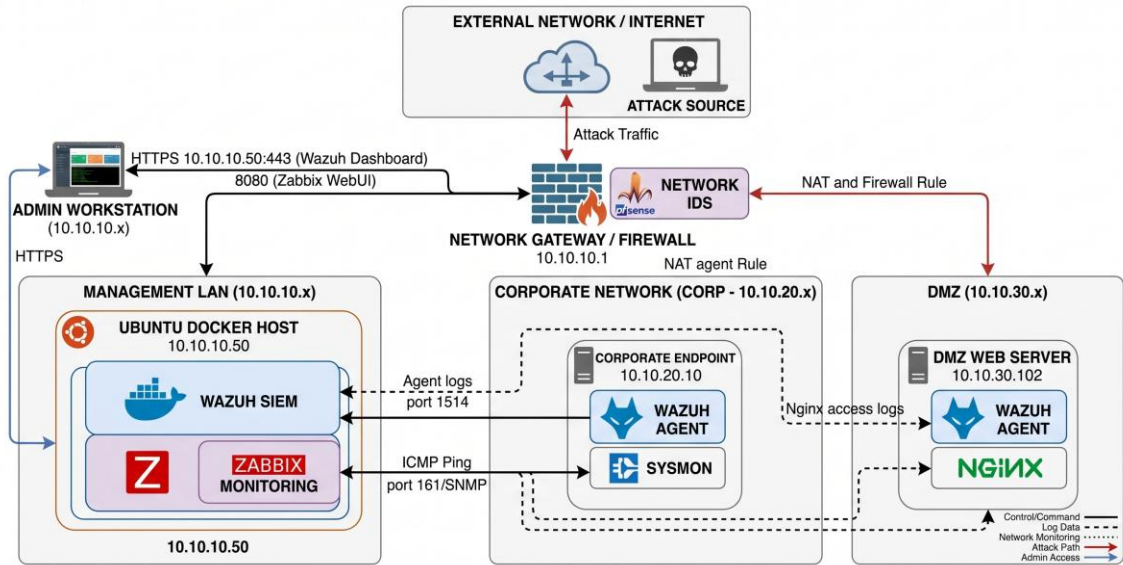
- Automated incident remediation (SOAR capabilities).
- High availability/failover clustering for the management servers.
- Full production deployment into a live corporate environment.

5. Architecture / How It Works

The network architecture is designed like a secure office building with different access zones. The pfSense firewall acts as the security guard at the front door and the hallways, ensuring traffic only flows where it is allowed. The network is divided into three main areas:

- Management LAN (10.10.10.x): The 'control room'. This houses the main server (Ubuntu running Docker at 10.10.10.50), which runs both the Wazuh SIEM (security monitor) and Zabbix (health monitor).
- Corporate Network (10.10.20.x): The 'employee workspace'. This contains a simulated employee workstation running Windows 10 (win10-client at 10.10.20.10).
- DMZ (Demilitarized Zone - 10.10.30.x): The 'public lobby'. This houses public-facing services like our web server (dmz-web01 at 10.10.30.102).

Comprehensive SOC/NOC Hybrid Architecture: Segmentation & Data Flow



Small programs called 'agents' are installed on the Windows and DMZ servers. These agents act like security cameras and health sensors, constantly sending data back to the control room. If a server stops responding (caught by Zabbix) or if someone tries to break in (caught by Wazuh), an alert is generated.

6. Tools & Technologies Used

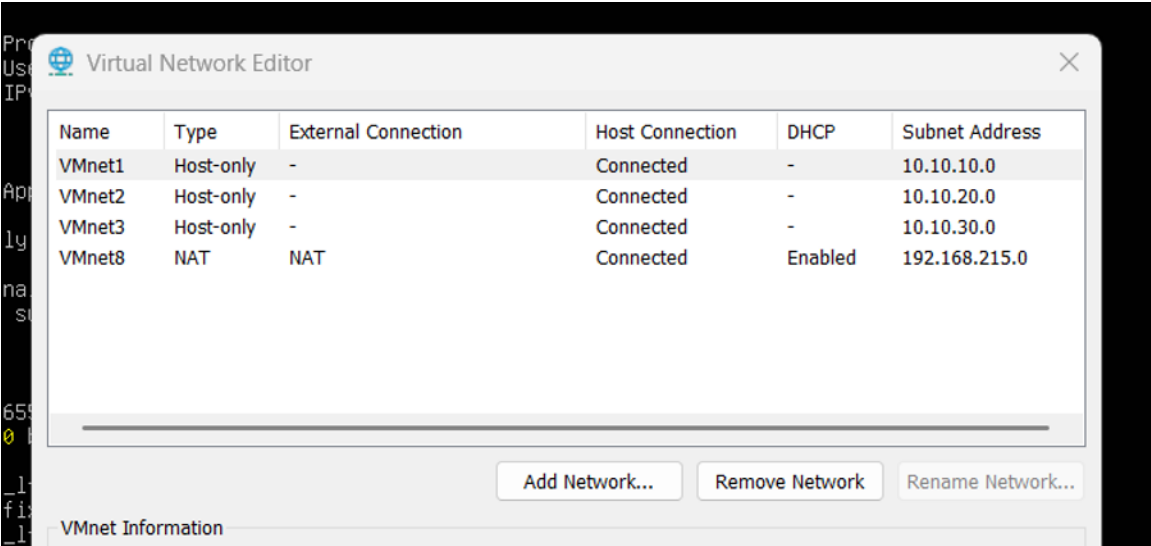
Tool / Platform	Function / Role	Reason for Selection
pfSense	Network routing, segmentation, and firewall boundary enforcement.	Industry-standard open-source firewall providing robust VLAN capabilities.
Suricata	Intrusion Detection System (IDS) at the network perimeter.	Provides deep packet inspection to catch malicious traffic before it hits the endpoints.
Wazuh	SOC platform for log analysis, file integrity monitoring, and endpoint detection and response (EDR).	Comprehensive, open-source SIEM that seamlessly maps alerts to the MITRE ATT&CK framework.
Zabbix	NOC platform for infrastructure and performance monitoring.	Highly scalable, supports both agent-based and agentless (SNMP/ICMP) monitoring.
Docker & Compose	Containerization platform to host Wazuh and Zabbix.	Ensures consistent, reliable deployment of complex backend systems on a single host.
Sysmon	Advanced Windows endpoint	Provides granular visibility

	telemetry.	into process creation and network connections on the Windows client.
Kali Linux	Offensive security testing platform.	Used to simulate real-world attacks (SQLi, XSS, Privilege Escalation) to validate defenses.

7. Setup & Installation Guide

Step 1: Network Preparation

1. Install pfSense and configure three separate network interfaces (LAN, CORP, DMZ).
2. Assign IP subnets to each interface and create firewall rules to allow management traffic from LAN to CORP/DMZ.



Interface Assignments
Interface Groups
Wireless
VLANs
QinQs
PPPs
GRES
GIFs
Bridges

LAGGs

Interface	Network port
WAN	em0 (00:0c:29:b3:93:c2) ▼
LAN	em1 (00:0c:29:b3:93:cc) ▼ Delete
CORP	em2 (00:0c:29:b3:93:d6) ▼ Delete
DMZ	em3 (00:0c:29:b3:93:e0) ▼ Delete

Save

Step 2: Core Server Deployment

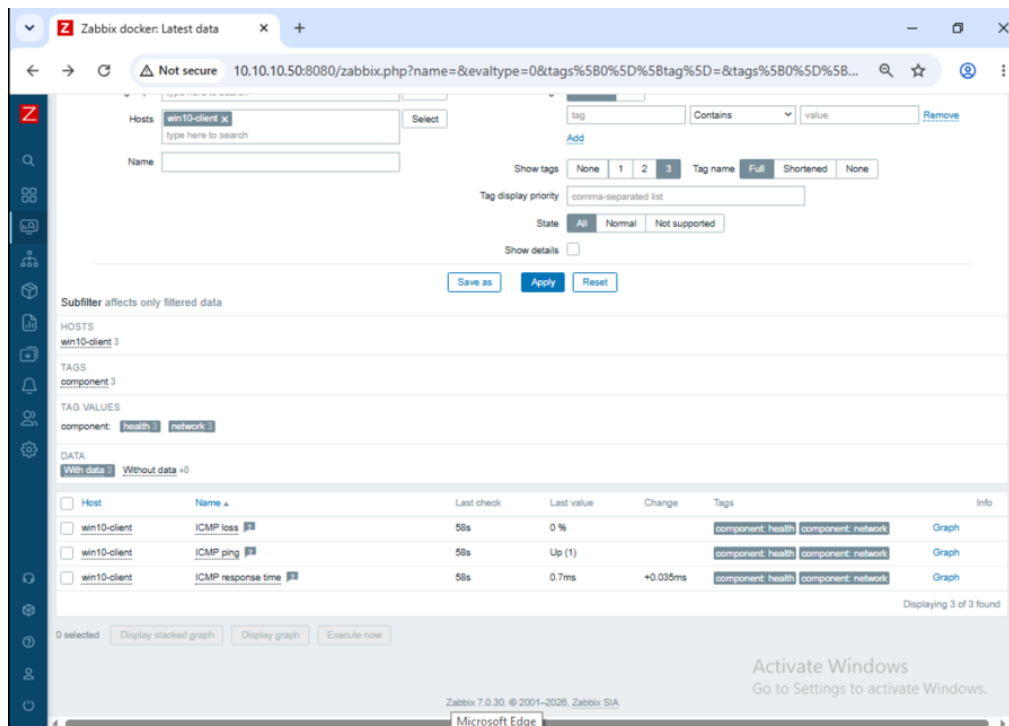
- Install Ubuntu Server on the LAN segment (IP: 10.10.10.50).
- Install Docker and Docker Compose.
- Deploy the Wazuh Manager/Indexer stack and Zabbix Server using their respective docker-compose.yml files.

Step 3: Endpoint Agent Installation

- Deploy a Windows 10 VM on the CORP network and an Ubuntu Server on the DMZ.
- Install the Wazuh Agent on both endpoints and connect them to the manager.
- Install Sysmon on the Windows 10 VM to enhance log generation.

Step 4: Monitoring Configuration

- In Zabbix, add pfSense and the DMZ server using SNMP templates.
- Add the Windows 10 client using an ICMP Ping template.
- Enable 'File and Printer Sharing (Echo Request - ICMPv4-In)' on the Windows Firewall to allow Zabbix pings.



8. Workflow / Day-to-Day Operations

The daily operation of this hybrid NOC/SOC environment involves:

- NOC Monitoring (Zabbix): Dashboards are reviewed to ensure all hosts show green availability statuses. Graphs for network interface traffic and ping response times are monitored for anomalies.

Zabbix docker: Hosts

10.10.10.50:8080/zabbix.php?name=&ip=&dns=&port=&status=-1&evaltype=0&tags%5B0%5D%5B...

Hosts

Create host

Name:

Host groups: type here to search

IP:

DNS:

Port:

Status:

Tags: tag Contains value

Show hosts in maintenance: ☒ Show suppressed problems: ☐

Severity: ☐ Not classified ☐ Warning ☐ High ☐ Information ☐ Average ☐ Disaster

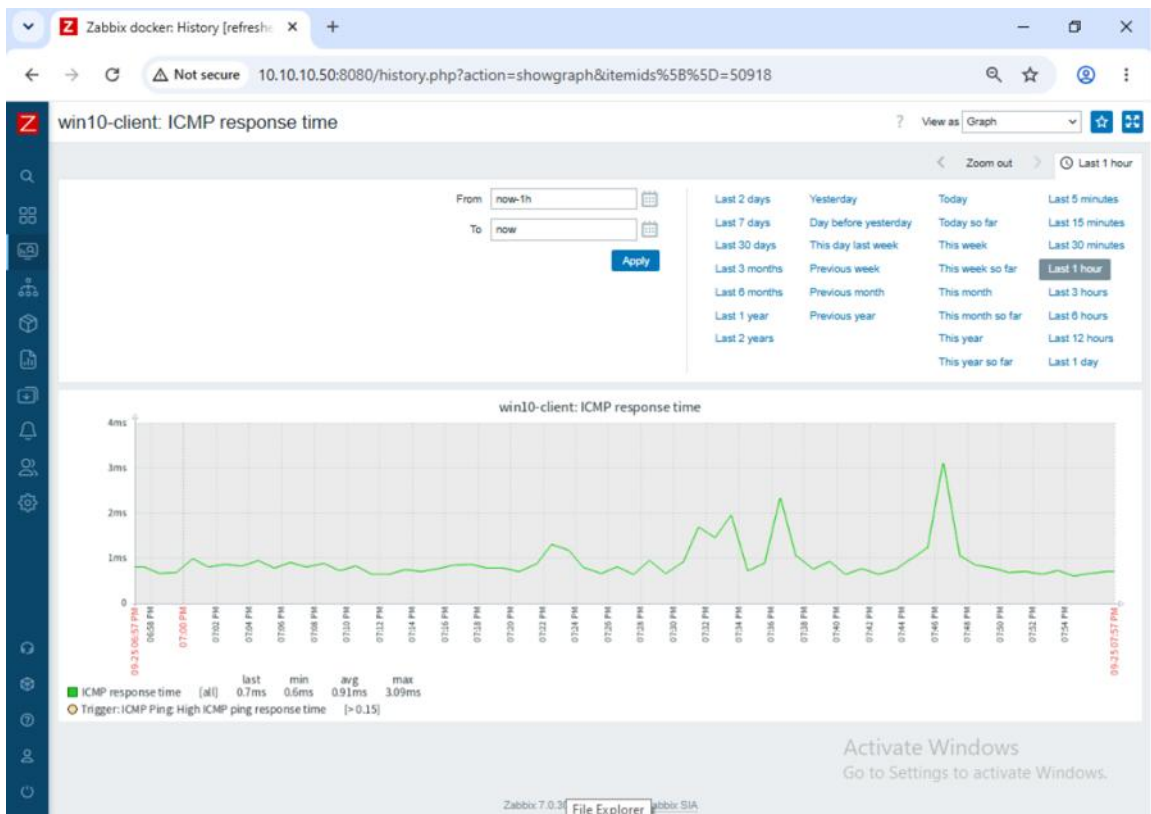
Name	Interface	Availability	Tags	Status	Latest data	Problems	Graphs	Dashboards	Web
dmz-web01	10.10.30.102:161	SNMP	class: os target: linux	Enabled	Latest data 57	Problems	Graphs 9	Dashboards 3	Web
pfSense-Gateway	10.10.10.1:161	SNMP	class: network target: generic	Enabled	Latest data 48	Problems	Graphs 4	Dashboards 1	Web
win10-client	10.10.20.10:10050	ZBX	class: network target: icmp	Enabled	Latest data 3	Problems	Graphs	Dashboards	Web
Zabbix server	127.0.0.1:10050	ZBX	class: os class: software target: linux ***	Enabled	Latest data 121	1	Graphs 8	Dashboards 4	Web

Displaying 4 of 4 found

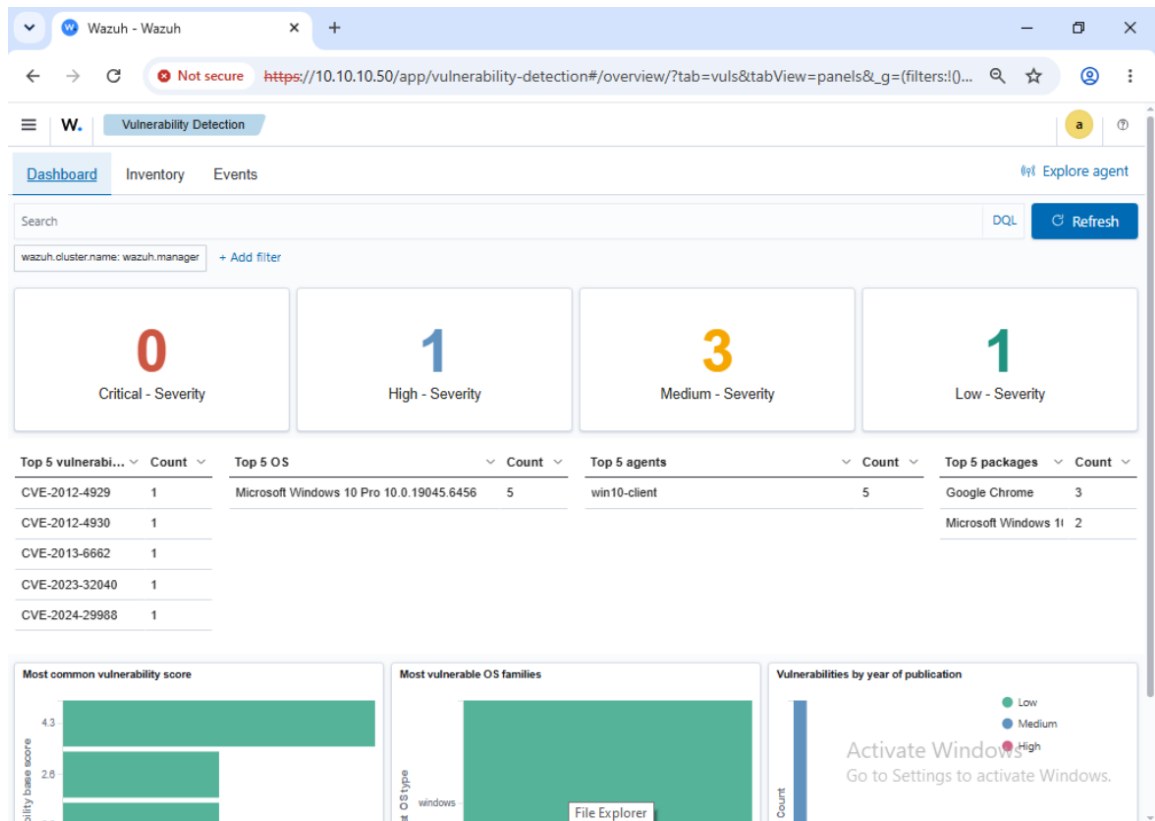
Activate Windows
Go to Settings to activate Windows.

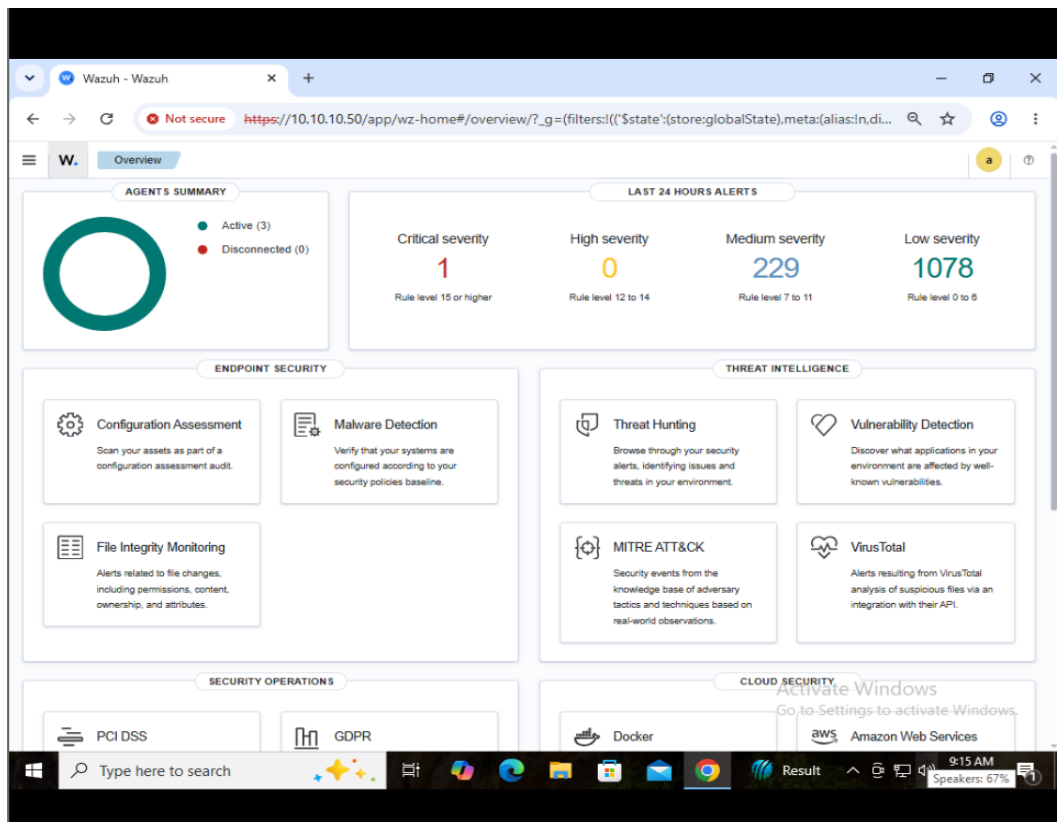
Zabbix 7.0.30, © 2001–2026, Zabbix SIA



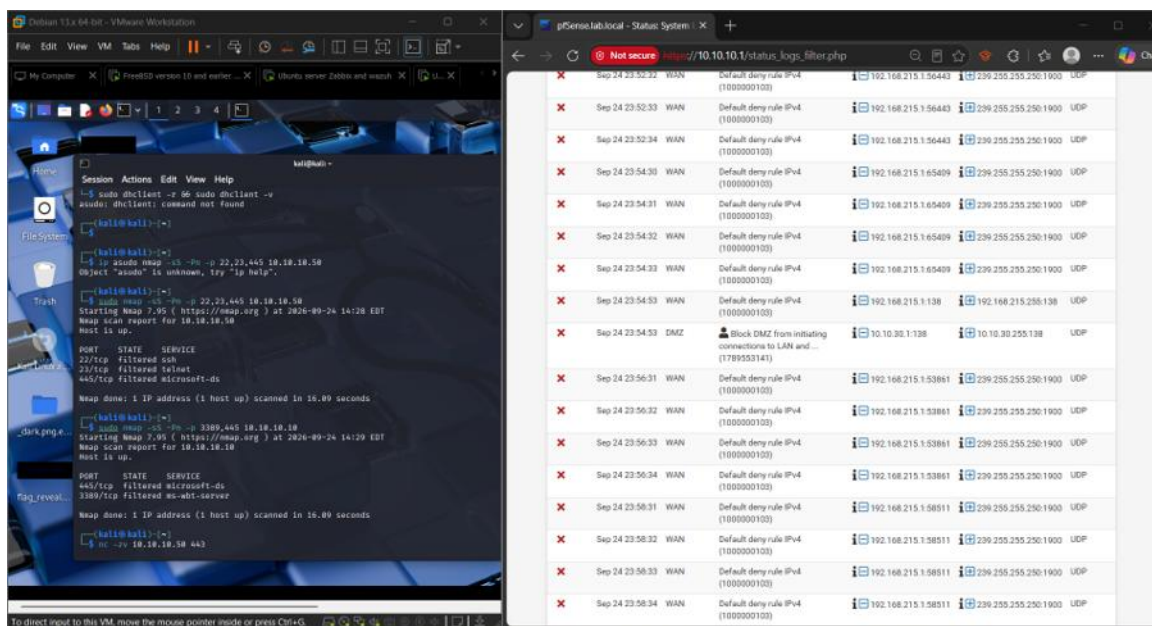


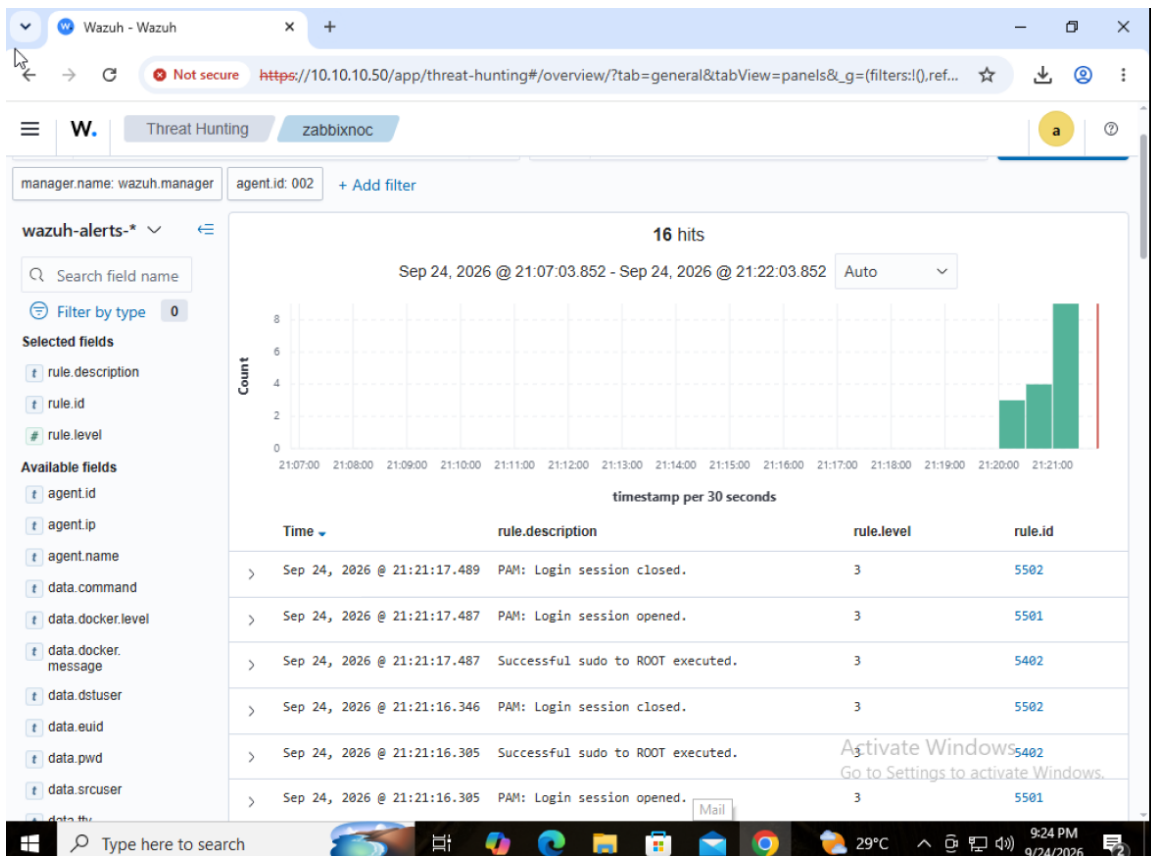
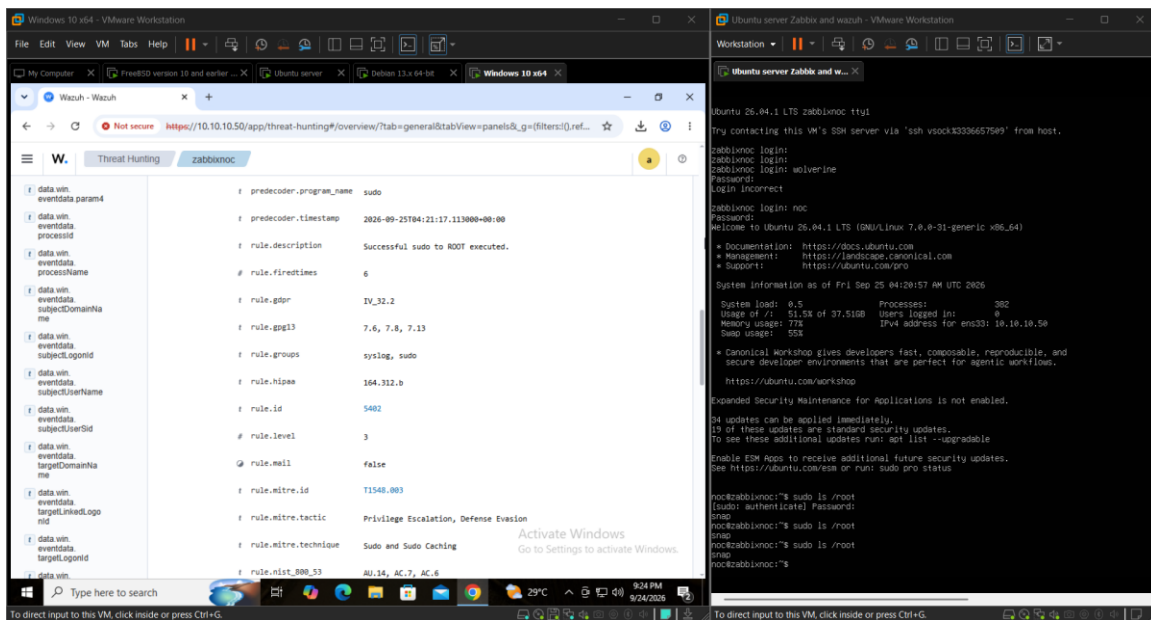
- SOC Monitoring (Wazuh): The SIEM dashboard is actively monitored for security alerts. Events are categorized by severity. Medium (Level 7-11) events are logged for review, while High (Level 12-14) and Critical (Level 15+) events trigger immediate incident response procedures.

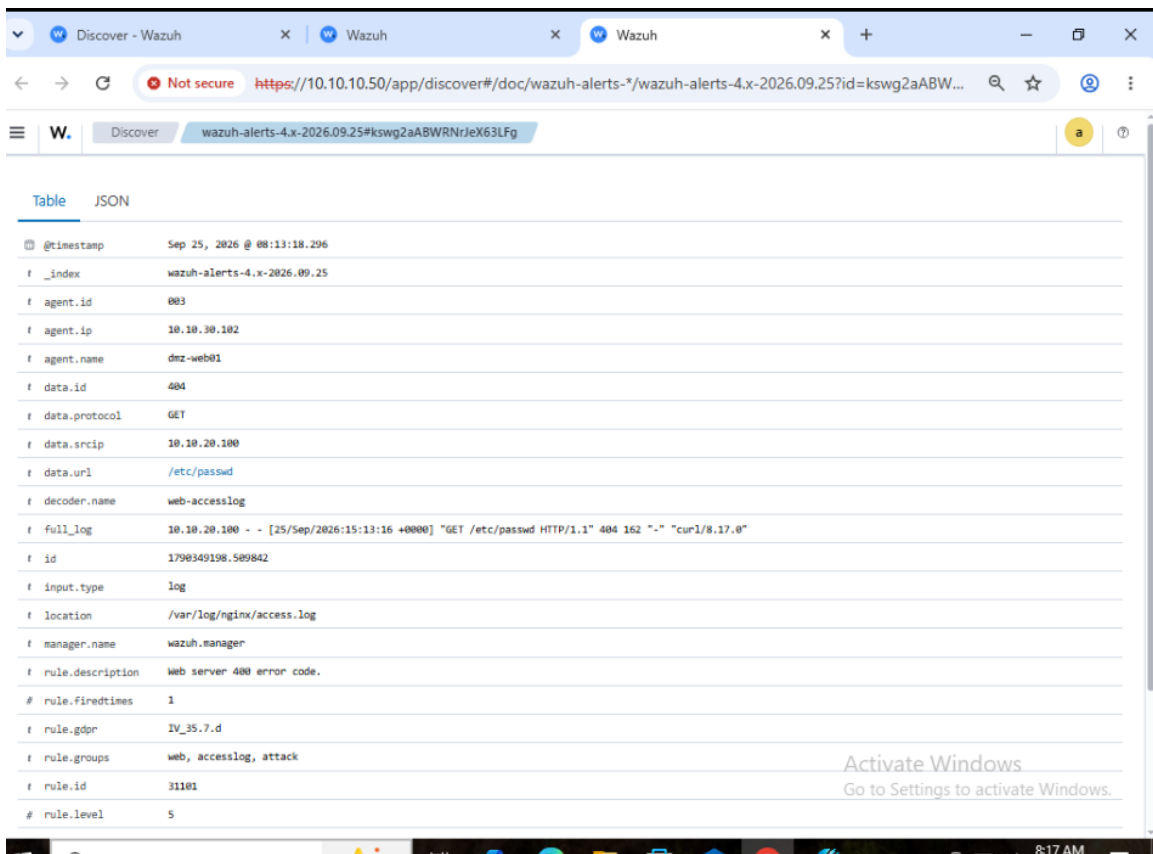
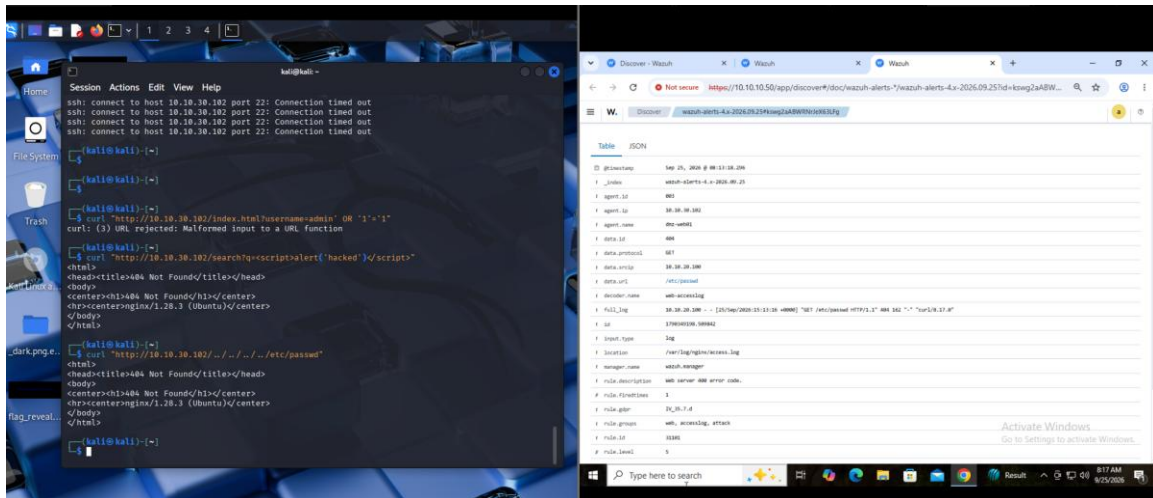


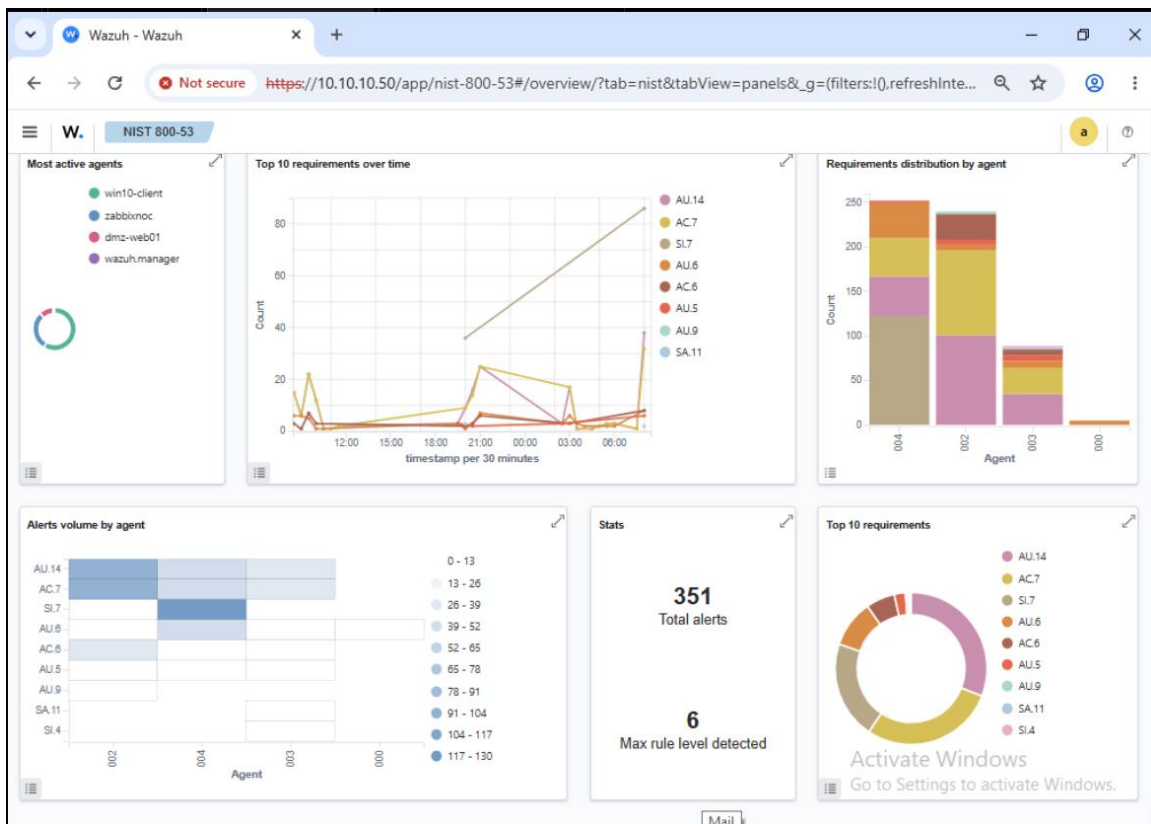
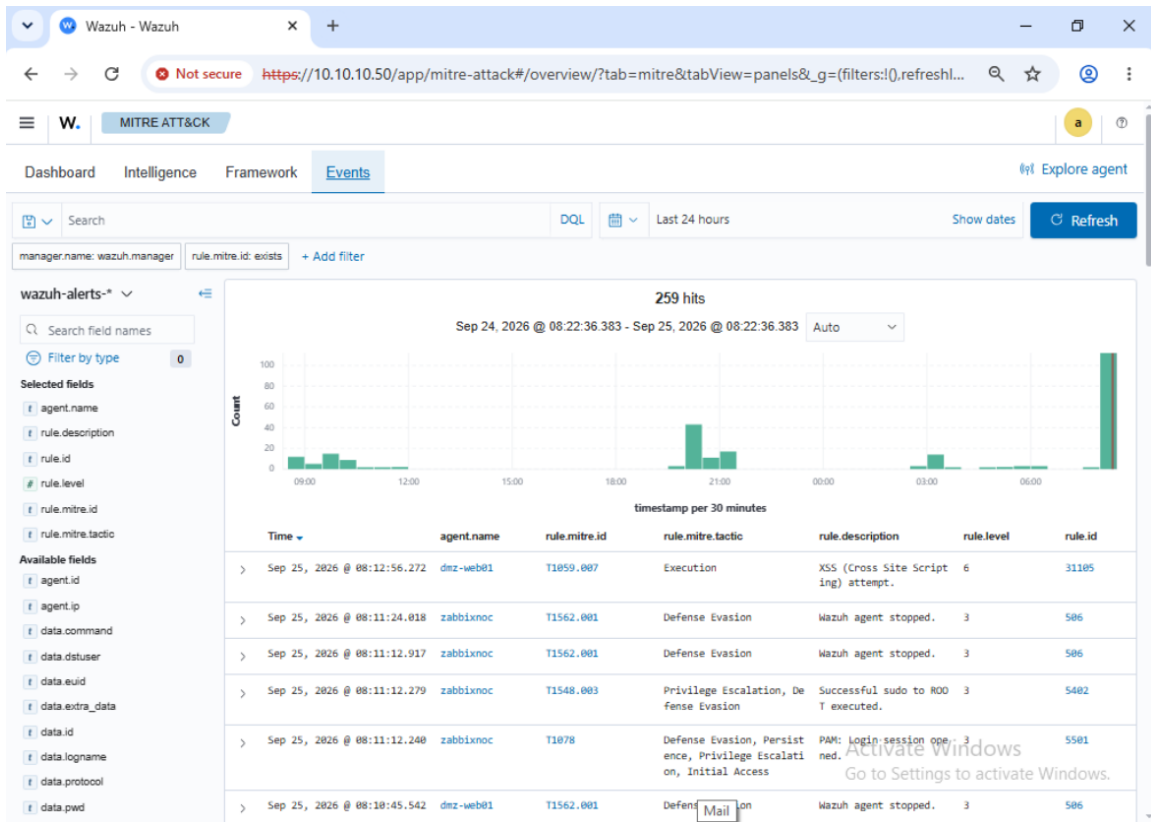


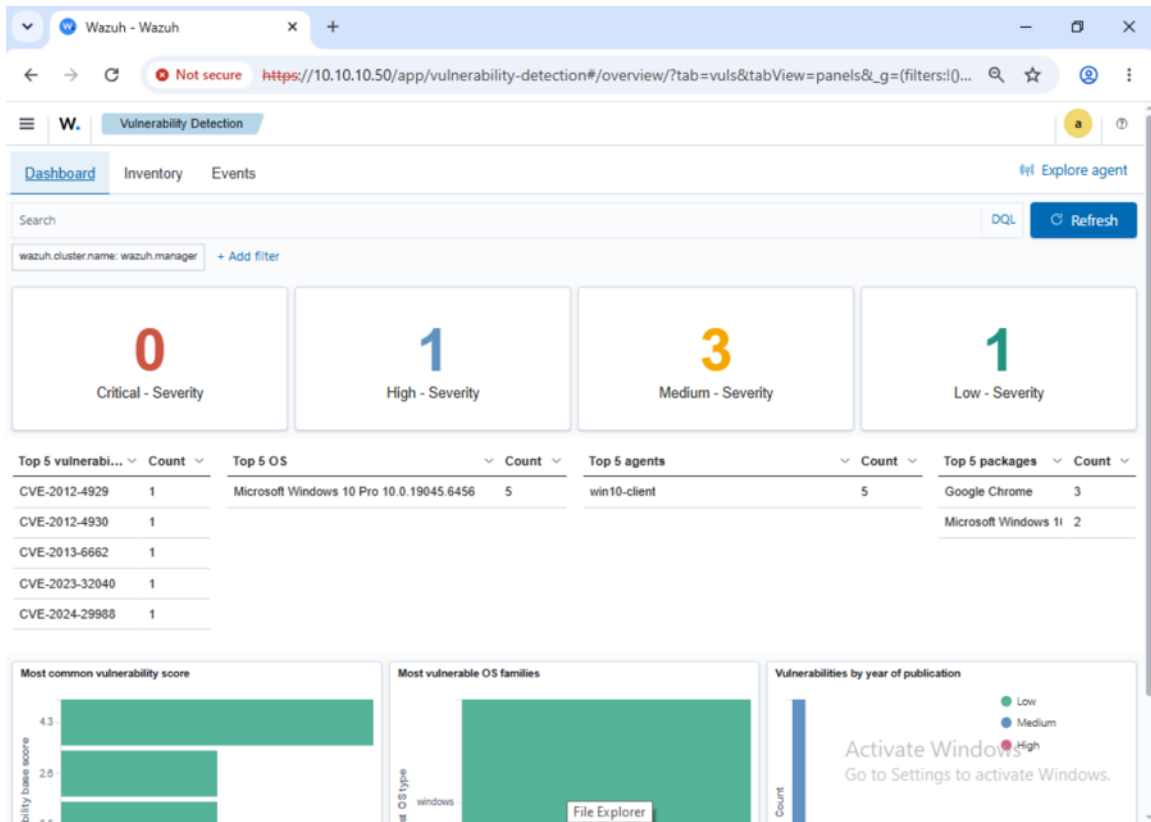
- Incident Handling: Following NIST SP 800-61 Rev 2 guidelines, when a threat (such as an OWASP web attack or sudo privilege escalation) is detected, the team identifies the source IP, contains the threat (e.g., via pfSense firewall block), eradicates the vulnerability, and documents the incident.









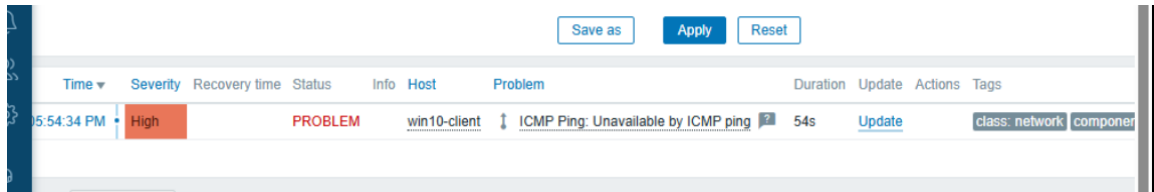


9. Challenges Faced & How They Were Solved

Challenge / Issue	Root Cause	Solution Applied
Wazuh OpenSearch 'Internal Server Error' (Null Pointer Exception)	The dashboard DQL search bar requires specific uppercase boolean operators (OR) and exact field matching for nested JSON objects.	Bypassed DQL syntax errors by using the built-in UI 'Add Filter' tool (filtering by 'rule.level >= 3').
Missing Alerts in Wazuh Dashboard (Empty Results)	Timezone clock drift (UTC in container vs Local Time in browser) and hard-pinned agent filters hiding recent events.	Cleared pinned filters, expanded time range to 'Last 24 hours', and used the raw 'Discover' tab. Verified backend ingestion directly via alerts.json.
Zabbix reporting 'Unavailable by ICMP ping' for Windows 10 Client	Built-in Windows Defender Firewall drops incoming ICMP (ping) requests by default, causing the health check to fail.	Ran a PowerShell command to enable the 'File and Printer Sharing (Echo Request - ICMPv4-In)' firewall rule.

10. Lessons Learned & Best Practices

- **Log Verification:** Always verify log ingestion at the backend (e.g., checking alerts.json in Wazuh) before assuming the dashboard UI is broken.
- **Syntax Matters:** When querying databases, precise syntax and understanding field mappings are critical to prevent interface crashes.
- **Default Firewall Rules:** Operating systems often have strict default firewalls (like Windows blocking ICMP) that can interfere with standard monitoring protocols.



The screenshot shows a table of alerts in the Wazuh dashboard. At the top, there are buttons for 'Save as', 'Apply', and 'Reset'. The table has columns for Time, Severity, Recovery time, Status, Info, Host, Problem, Duration, Update, Actions, and Tags. A single row is visible with the following data: Time: 5:54:34 PM, Severity: High (in an orange box), Status: PROBLEM (in a red box), Host: win10-client, Problem: ICMP Ping: Unavailable by ICMP ping (with a help icon), Duration: 54s, Update: (link), Actions: Update (link), and Tags: class: network, componer (partially visible).

Time	Severity	Recovery time	Status	Info	Host	Problem	Duration	Update	Actions	Tags
5:54:34 PM	High		PROBLEM		win10-client	ICMP Ping: Unavailable by ICMP ping ?	54s	Update	Update	class: network componer

11. Future Improvements / Roadmap

- Implementation of automated Active Response in Wazuh to block attacking IPs automatically.
- Integration of a vulnerable web application (like OWASP Juice Shop) on the DMZ to test more complex web attack vectors.
- Configuration of email or Slack integrations to alert administrators of critical Zabbix or Wazuh events in real-time.

12. Glossary

NOC (Network Operations Center): A centralized location where IT administrators manage, control, and monitor network health and performance.

SOC (Security Operations Center): A centralized function where a team continuously monitors and analyzes an organization's security posture.

SIEM (Security Information and Event Management): Software that gathers and analyzes activity from resources across an IT infrastructure.

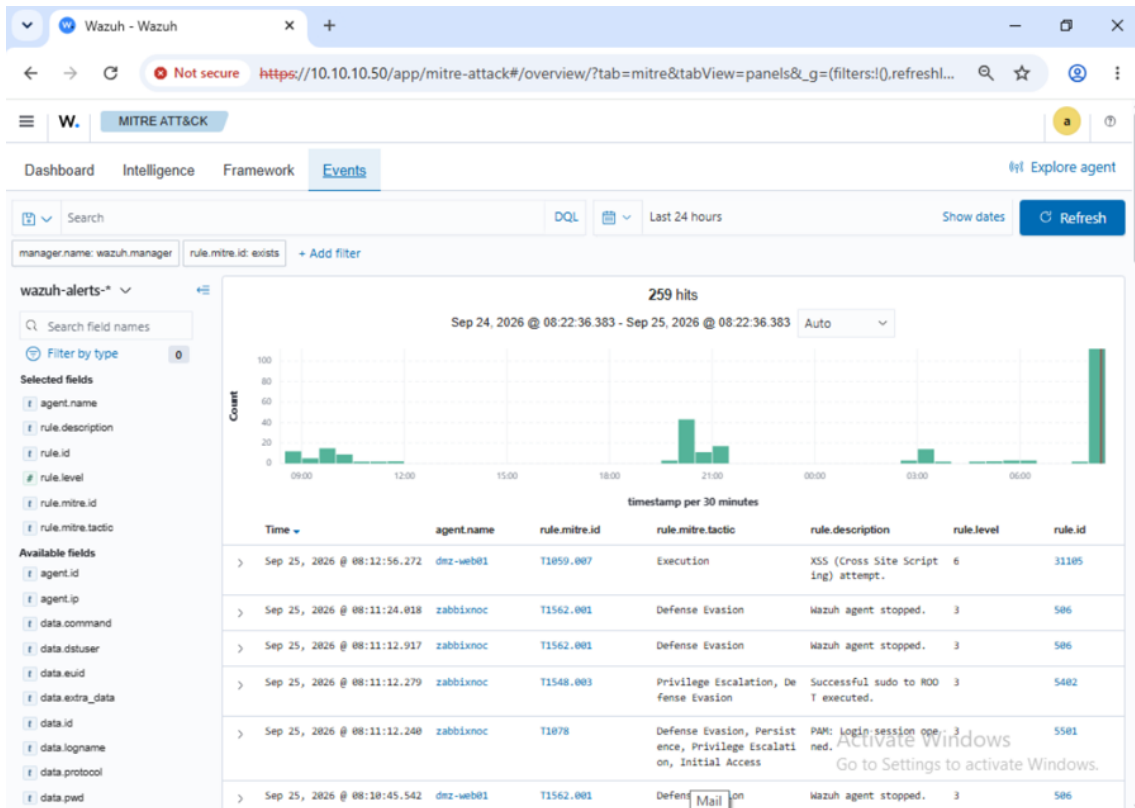
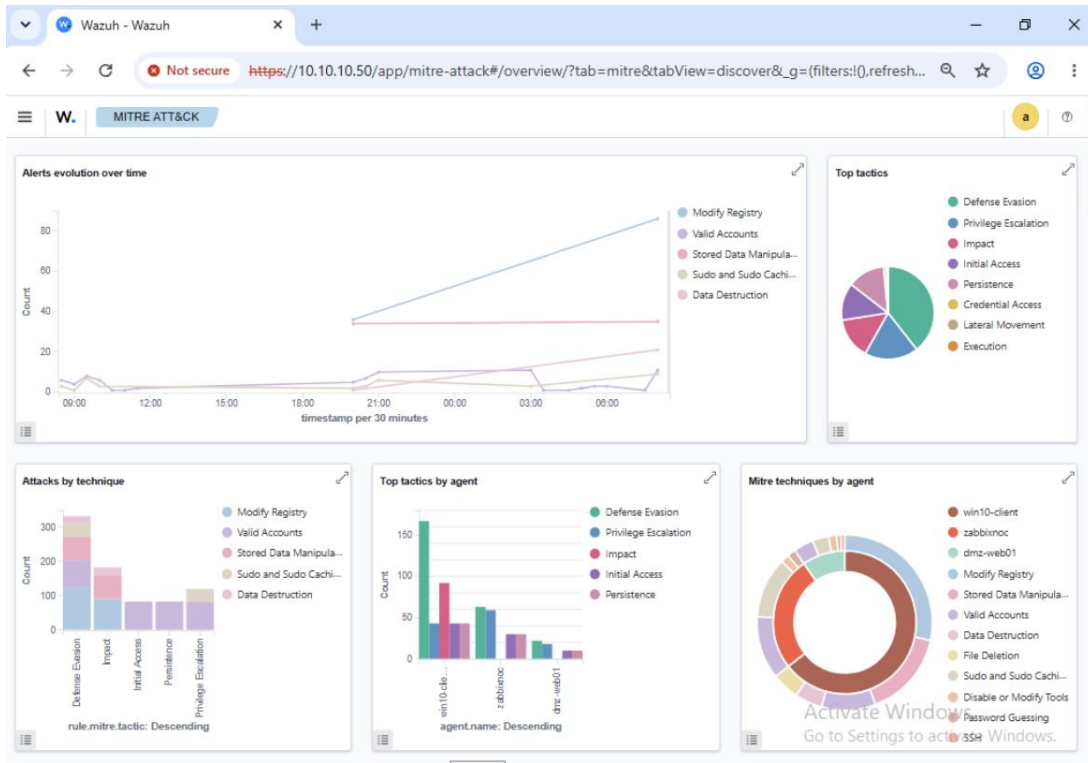
VLAN (Virtual Local Area Network): A custom network created from one or more existing LANs.

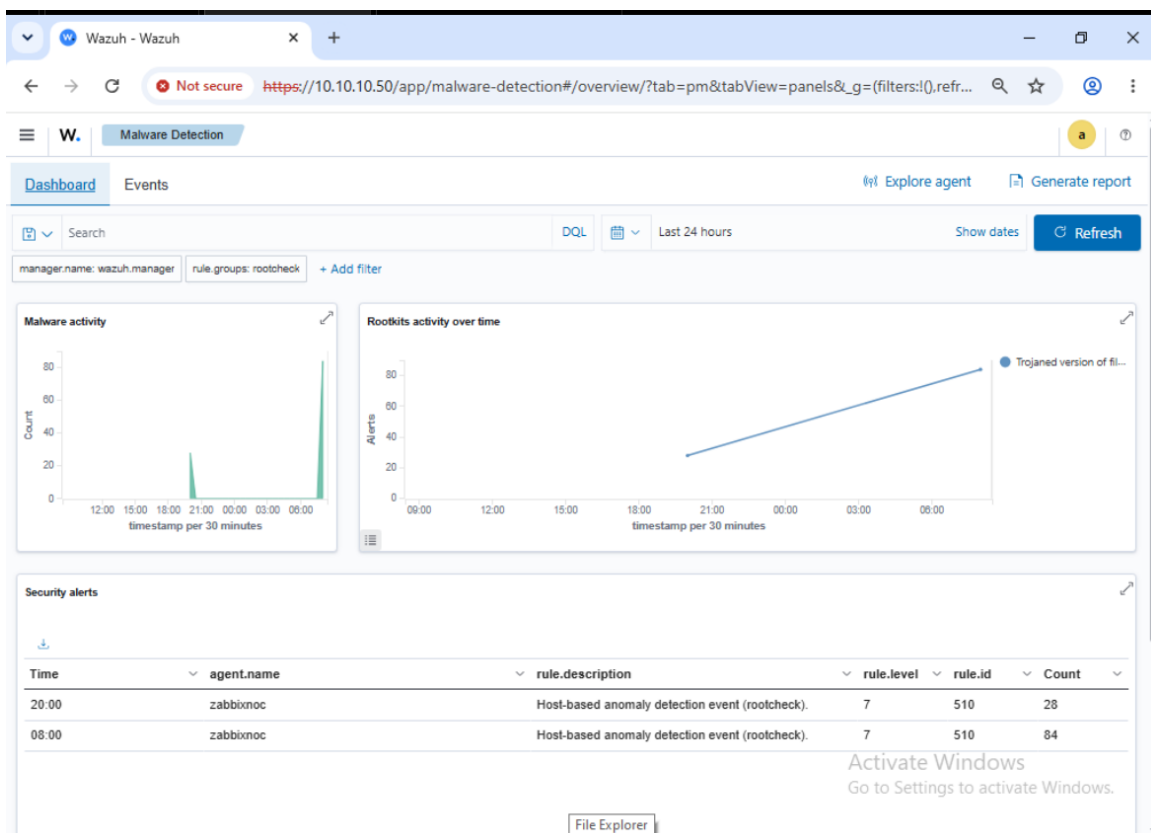
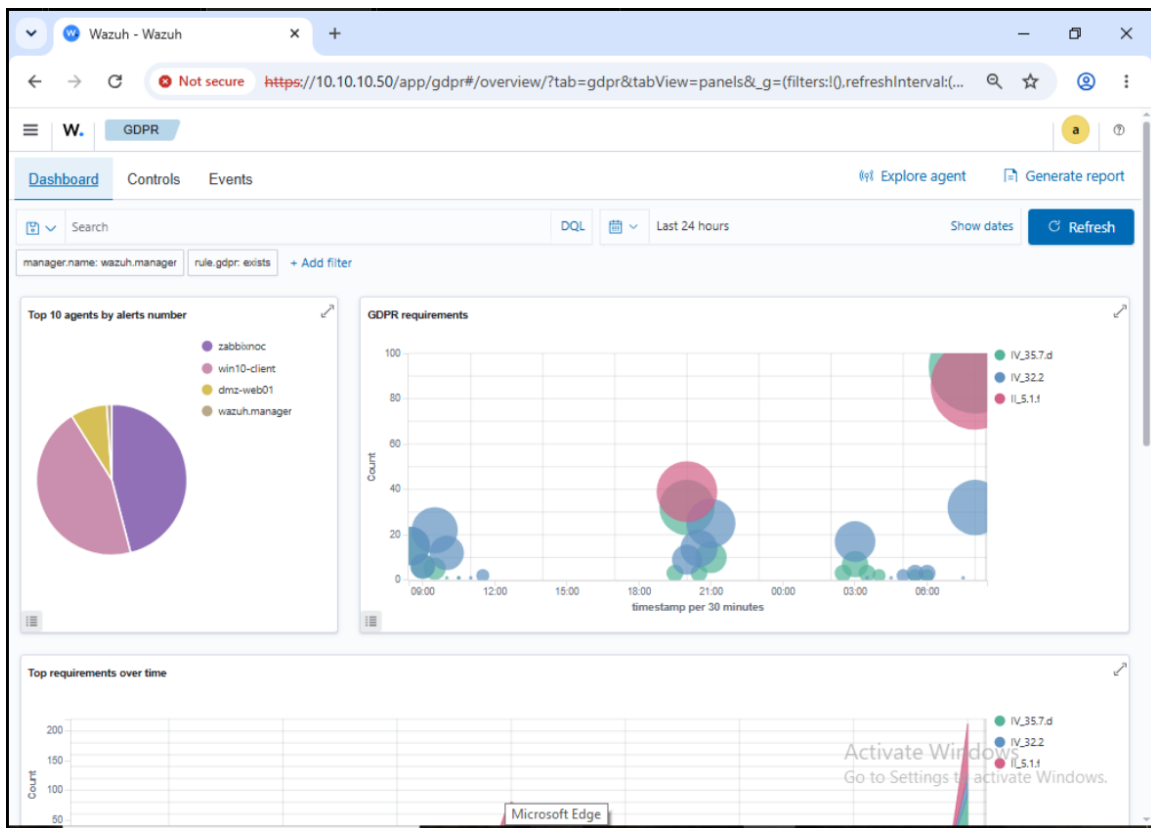
MITRE ATT&CK: A globally-accessible knowledge base of adversary tactics and techniques.

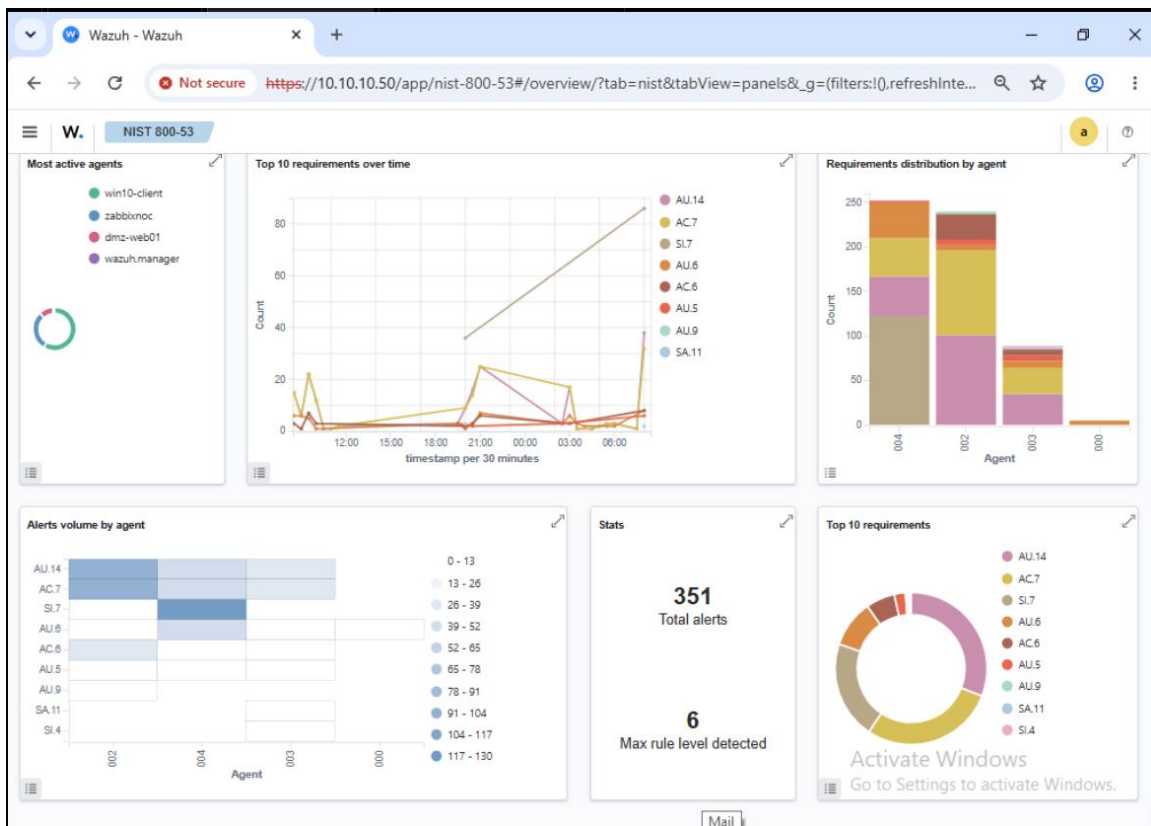
ICMP (Internet Control Message Protocol): A network layer protocol used to diagnose communication issues (e.g., 'ping').

SNMP (Simple Network Management Protocol): An Internet Standard protocol for collecting information about managed network devices.

13. Appendix







14. Info Needed

The following details were not explicitly detailed during the troubleshooting sessions and may need to be filled in manually to complete the documentation:

- Specific hardware specifications (CPU, RAM, Storage) allocated to each VM/Container.
- Exact Suricata IDS configuration steps and rule sets deployed.
- Details on specific NIST SP 800-61 Rev 2 incident response playbooks customized for this lab.